The Physics of Artificial Intelligence

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Preface

Actually, I am not an AI or CS student. When the AI explosion occurred 3 years ago, I did not care about it so much. I thought it was a scam; created by the billionaires, who want to take control of the new world. I was not interested in AI in the first place. Everything was nonsense and meaningless, why did I have to care about it?

Like every student living in AI-era, I was forced to learn a subject called "Fundamentals of Artificial Intelligence", 3 credits, in my third year. When the midterm came, I realized that I did not know anything; and the risk of failing this subject was real. 2 hours before midterm, I came to the library and tried to learn something that could save my grade above 4 points. To be honest, the feeling of just studying to merely pass a stupid test and then being done with it was so frustrating. I was watching many tutorial videos on YouTube, and thought "OK, the game is over. I am done.". My brain could not consume any knowledge from them. I was a failure.

Instead of learning like crazy in the last 2 hours left, I just skimmed all the stuffs for 1 hour, and decided to give up. Fortunately, I did misclick a video; this moment would change my vision forever.

"Geoffrey Hinton, Nobel Prize in Physics 2024: Banquet speech"

Wait, did something go wrong here? Nobel Prize in Physics for Artificial Intelligence? My brain turned on because I deeply cared about Physics. "Why did something bullshit receive a Nobel Prize, even in Physics?". I did some searching, and then I found some familiar keywords, such as "Boltzmann Statistics", "Spin-Glass", "Ising Model". "Perhaps I could do some Physics here.", I thought. Time flies so fast, half a year has already passed. I am still confused, but I think I have got an idea. After my first book "Thinking like a Frequentist" was published online, why not try writing about the "bridge" between Physics and AI? At least so far, I have not seen anyone write a sufficiently understandable document on this topic. But the problem is, I am not even a Physics student. Up to now, my Physics knowledge is limited to Halliday textbooks, and some experience with digital and analog electronics. To fill these gaps, I have to learn everything from scratch.

That is why I wrote this book as a concise self-study guide for anyone interested in Physics and AI. Unlike my first book, which was written when my knowledge of Probability and Statistics was already solid; this book was written during my self-study process. I may have made many mistakes or misinterpretations; but I hope you, my readers, can point them out and send feedbacks via email `tinvu1309@gmail.com`. I would be extremely happy if someone could discuss these issues with me.

The structure of this book contains 2 main parts: "Introduction to Statistical Physics" from Chapter 0 to Chapter 7, and "The Physics of Artificial Intelligence" from Chapter 8 to Chapter 15. Although I will try to keep everything as simple as possible; but before reading, you should be familiar with elementary university mathematics concepts, include: *Calculus*, *Linear Algebra* and *Probability and Statistics*. The better your math skills, the higher your physics ability. In physics, you need to understand some basic concepts of *Mechanics and Thermodynamics*; *Electromagnetism* and some applied physics subjects such as *Signals and Systems*, *Control Theory* and *Digital Signal Processing*. Some useful books and lecture notes are listed below:

1. Calculus: Calculus 7E, James Stewart
2. Linear Algebra: Elementary Linear Algebra, Larson - Edwards - Falvo
3. Probability and Statistics: Probability and Statistics for Engineers and Scientists, Walpole - Myers or Thinking like a Frequentist, Tin Vu
4. Mechanics and Thermodynamics, Electromagnetism: Fundamentals of Physics, Halliday and Resnick
5. Signals and Systems: Signals and Systems, Alan V. Oppenheim or my lecture note (Vietnamese)
6. Digital Signals Processing: My lecture note (Vietnamese)
7. Control Theory: Control Systems Engineering, Norman S. Nise or my lecture note (English)

I highly suggest that you should read the first part **without any AI tools** to build physics intuition (I did that too); before diving into the second part of this book, where we "wire" our intuition up with the aid of AI tools. Everything will turn out great as long as we keep trying.

Part I

Introduction to Statistical Physics

Chapter 0

The Zeroth Law of Thermodynamics

0.1 What is Temperature?

If you touch an ice cube, you will feel cold; and if you touch boiling water, you will feel hot. Now you can conclude that ice cube is at low temperature, and boiling water is at high temperature. In common thinking, **temperature** is a quantity that used to measure hotness and coldness of objects. Instead of trusting our feeling, we often use a measuring device called **thermometer** to estimate the temperature.

Example 0.1.1. Using thermometer to measure the temperature of a cup of hot water.

After adjusting the thermometer to normal room temperature, immerse it into a cup; and wait until the mercury inside no longer rises. Now read the temperature displayed on the thermometer.
The waiting time is called **relaxation time**; the state when both thermometer and hot water sharing the same temperature is called **thermal equilibrium**.

0.2 The Zeroth Law of Thermodynamics

Theorem 0.2.1 (The Zeroth Law of Thermodynamics)**.** *If object A and B are each in thermal equilibrium with a third object T, then A and B are in* ***thermal equilibrium*** *with each other.*

In spite of its simplicity, this law provides an independent definition of temperature without reference to any complex terminologies.

Corollary 0.2.1.1 (The First Definition of Temperature)**.** *Every object has a property called* ***temperature****. When two objects are in* ***thermal equilibrium****, their temperatures are* ***equal****.*

After reading the temperature of hot water in our cup (might be around 60°C), you wait for a while, and the mercury level inside thermometer slowly decreases. You carefully observe until the water has completely cooled down to room temperature. Finally, the mercury level stops decreasing, and at this point the room temperature, the water temperature, and the thermometer temperature are all the same (possibly 25°C), according to the law.

0.3 Thermal Expansion and Heat Absorption

Let's look closer to the experiment that we did. If I use a clock to measure time, I will obtain a simple graph:

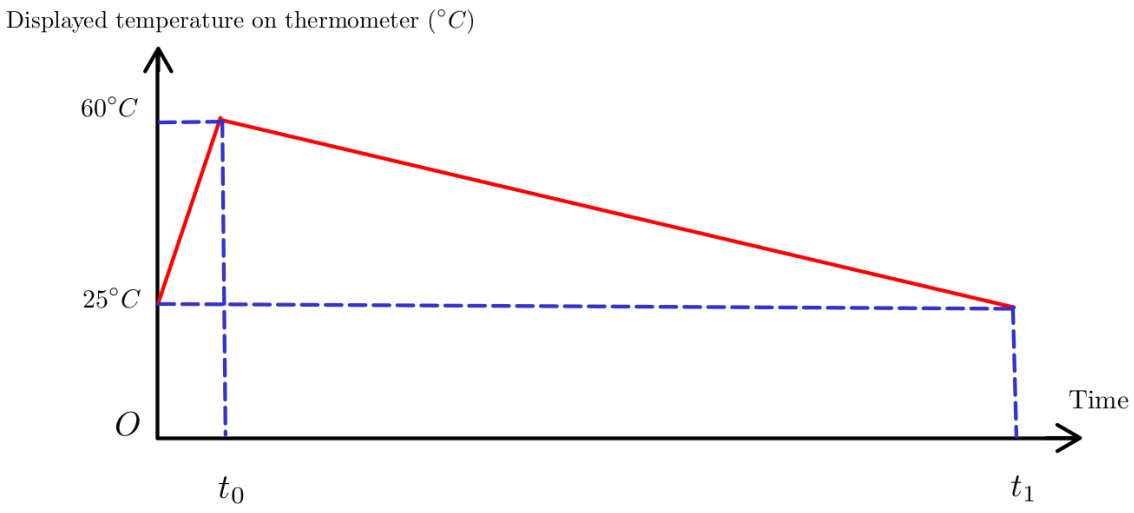

Figure 1: Relationship between time and thermometer's temperature

As you can see here, the time interval $(0, t_0)$ is very short compared to (t_0, t_1). Look at the first interval [1], why did the thermometer's temperature increase?

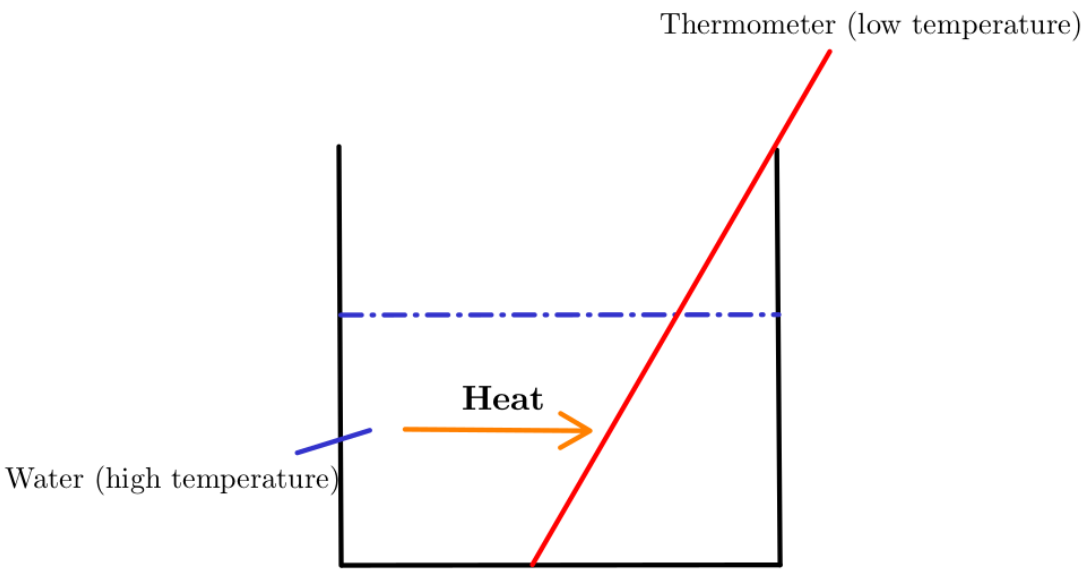

Figure 2: Heat flows between objects

The simplest explanation is that there is a "special energy" that transfers from a high-temperature object to a low-temperature object. This is very easy to imagine, and we call that "energy" as **heat** [2]. Now you might wonder "Why does not heat flow in the reverse direction?". This question will be explained carefully in the next chapters.

Definition 0.3.1 (Heat Definition)**.** *Heat is the energy that* ***spontaneously*** *transfers from the higher temperature object to the lower temperature object.*

[1] For a very short period, the amount of energy transferred to the outside environment is negligible, so we can ignore it.

[2] **Heat** in this context is a **positive quantity**. I will explain the negative case later.

Because heat is a **transferring energy** between objects, so we have to be careful here:

Corollary 0.3.0.1. *A thermodynamics system **does not** contain heat.*

Now we can form the second definition of **temperature** bases on Definition 0.3.1 as follows:

Corollary 0.3.0.2 (The Second Definition of Temperature)**.** ***Temperature** is a measure of the tendency of an object to **spontaneously** give up energy to its surroundings.*

Obviously, from Definition 0.3.1 and Corollary 0.3.0.2, if we define Q as heat and T as temperature, we obtain the proportional relationship:

$$Q \propto \Delta T$$

Definition 0.3.2 (Heat Capacity Definition)**.** *The **heat capacity** C [1] of an object is defined as:*

$$Q = C\Delta T = C(T_f - T_i)$$

As you can see here, the unit of quantity C is (J/K)[2]. In practice, it is more convenient to define a new quantity as "heat capacity per mass" c (J/K.kg), and standardize their values into a useful look-up table for searching.

Corollary 0.3.0.3 (Specific Heat Definition)**.** *The specific heat is defined as:*

$$c = \frac{C}{m}$$

Or equivalent to:

$$Q = mc\Delta T$$

Since the initial temperature of hot water is higher than thermometer, hot water tends to give up energy in **heat form** to the thermometer. We can easily calculate how much released energy using the above formula.

Example 0.3.1. How much heat flows from hot water to thermometer? Known that $m_{\text{Hg}} = 0.5$ (g) and $c_{\text{Hg}} = 138$ (J/K.kg).

We have:

$$Q = mc\Delta T^3 = \frac{0.5}{1000}.138.(60 - 25) = 2.415 \text{ (J)}$$

Example 0.3.2. How much heat flows from the thermometer to hot water? What is the temperature of hot water at the moment t_0? Known that $m_{\text{H}_2\text{O}} = 50$ (g) and $c_{\text{H}_2\text{O}} = 4181$ (J/K.kg).

We have to very be careful here. Hot water gives **positive** energy in heat form to thermometer, so it receives **negative** energy flows from thermometer. Formally, we easily represent this relationship:

$$Q_{\text{hot water to thermometer}} = -Q_{\text{thermometer to hot water}}$$

The direction of heat flowing is very significant, that directly rules the **sign** of Q. Eventually, we obtain:

$$Q_{\text{thermometer to hot water}} = -2.415 \text{ (J)}$$

[1] Actually, I think the word "capacity" here is used due to historical reason, and it is not so accurate in physics context.

[2] Up to now, we still not define what K is; but you can think K (Kelvin) as a new temperature scale, like Celsius. The relationship between them is linear: $T_K = T_C + 273°\text{C}$.

[3] Converting Celsius to Kelvin scale does not affect ΔT.

Applying Corollary 0.3.0.3, we see that:

$$Q = mc\Delta T = \frac{50}{1000}.4181.(T_f - 60) = -2.415 \Rightarrow T_f = 59.98^\circ \text{ C}$$

Because ΔT is very low, so we can neglect this tiny changing and shifting our focus on the temperature changing of thermometer.
During the heat absorbing process, the mercury level slowly increases inside our thermometer, but why? Our common belief is that "If something is hot, it will get bigger, and vice versa.". Representing our claim into formulas, we obtain:

$$\Delta L \propto \Delta T$$

$$\Delta V \propto \Delta T$$

with L stands for length, and V stands for volume. In this chapter, we will temporarily accept our assertions (or common beliefs) as true without further explanation. They will be rediscovered in more detail in later chapters.

Definition 0.3.3 (Definition of Linear Expansion)**.** *If the temperature of a metal rod of length L is raised by an amount of ΔT:*

$$\Delta L = L\alpha\Delta T$$

The constant α is called the ***coefficient of linear expansion****.*

Using the same line of thinking, we can extend our Definition 0.3.3 above to volume expansion as a corollary.

Corollary 0.3.0.4 (Definition of Volume Expansion)**.** *If the temperature of solids, or liquids of volume V is raised by an amount of ΔT:*

$$\Delta V = V\beta\Delta T$$

The constant β is called the ***coefficient of volume expansion****.*

Obviously, the relationship between α and β can simply be described as:

Corollary 0.3.0.5.

$$\beta = 3\alpha$$

Both α and β share the same unit 1/K or K^{-1}, and like the specific heat quantity c, these coefficients are found by experiment.

Example 0.3.3. During the first time interval, the mercury temperature slowly increase from $T_i = 25°\text{C}$ to $T_f = 60°\text{C}$. Known that $\beta = 1.8.10^{-4}$ (K^{-1}), and the initial height of mercury level at T_i is $H_i = 10$ (cm), what is the height of mercury level at the moment t_0?

Notice that $V = \pi r^2 H \Rightarrow V \propto H$. Directly using the formula, we see that:

$$\Delta V = V\beta\Delta T \Rightarrow \Delta H = H\beta(T_f - T_i) = \frac{10}{100}.1.8.10^{-4}(60 - 25) = 6.3.10^{-4} \text{ (m)} = 0.063 \text{ (cm)}$$

The final height of mercury level is:

$$H_f = H_i + \Delta H = 10 + 0.063 = 10.063 \text{ (cm)}$$

To conclude this chapter, I would like to focus on a very special case, in which the heat capacity $C = +\infty$. This normally happens at a **phase transformation**, such as melting ice or boiling water.

$$C = \frac{Q}{\Delta T} = \frac{Q}{0} = +\infty$$

We still might want to know how much heat is required to melt or boil the substance completely during a phase transformation. This amount of energy is called the **latent heat**, denoted as L. Similar to the definition of specific heat, we define the quantity **specific latent heat**, denoted by l:

$$l = \frac{L}{m} = \frac{Q}{m} \text{ (J/kg)}$$

Example 0.3.4. How much ice should be added to a cup of 200 grams of boiling water to bring it down to 65°C. Assume that the ice is initially at −15°C, specific heat capacity of ice is 2.1 J/K.g, of water is 4.184 J/K.g; specific latent heat for melting ice is 333 J/g[1].

The flow of heat between ice and boiling water can be represented by formula:

$$Q_{\text{water}\to\text{ice}} = -Q_{\text{ice}\to\text{water}}$$

Or by the **law of conservation of energy** form:

$$Q_{\text{water}\to\text{ice}} + Q_{\text{ice}\to\text{water}} = 0 \text{ (internal energy is conserved, no external energy exists)}$$

Because 65°C is much higher than the freeze point of ice; when the system is equilibrium, all ice is **melted**.

$$\begin{aligned}
Q_{\text{water}\to\text{ice}} &= -Q_{\text{ice}\to\text{water}} \\
m_{\text{H}_2\text{O}} c_{\text{H}_2\text{O}} \Delta T_{\text{H}_2\text{O}} &= -(m_{\text{ice}} c_{\text{ice}} \Delta T_{\text{ice}} + m_{\text{ice}} l + m_{\text{ice}} c_{\text{H}_2\text{O}} \Delta T_{\text{melted ice raising temperature}}) \\
200.4.184.(65-100) &= -[m_{\text{ice}}.2.1.(0-(-15)) + m_{\text{ice}}.333 + m_{\text{ice}}.4.184.(65-0)] \\
\Rightarrow m_{\text{ice}} &= 46.01 \text{ (g)}
\end{aligned}$$

[1]In practice, instead of using standard SI units, we often use their variances like J/g, cal/g,... Chemists really love using gram.

Chapter 1

The First Law of Thermodynamics

1.1 The Microscopic Model

As you can see here, we are moving further than the zeroth law of thermodynamics. In the previous chapter, we neither know what is inside the substances, nor the essence of heat. Thinking like a physicist, we need to build a physical model to explain many phenomena in nature.

Substance is just the collection of many very tiny "imaginary" particles.

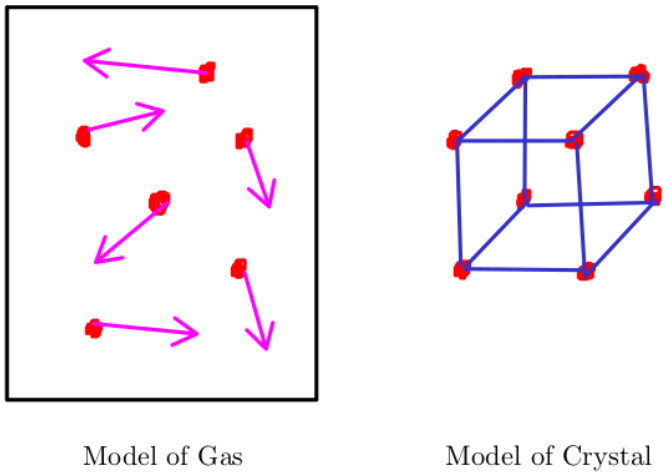

Figure 1.1: Our hypothesis model of gas and crystal

But why do I call these particles "imaginary"? Because in the 19th century, scientists had no way of directly observing them. Physicists living that era built the entire foundation of thermodynamics on atomic theory, which may not have convinced everyone until the quantum age. In the early 20th century, the existence of atoms was finally proven and widely accepted, supported by numerous undeniable physical and mathematical proofs. Finally, in 1955, the first photograph of atom was taken by Erwin W.Muller and his student.

Theorem 1.1.1 (Atomic Theory)**.** *Atomic theory is the specific theory that matter is composed of particles called* ***atoms****.*

I think Theorem 1.1.1 should be called axiom, in mathematical language. In the following chapters, we will focus in depth on how this theory can be applied to build the foundations of thermodynamics, rather than proving it correctness.

1.2 Volume and Pressure

I think everyone knows what volume is, but it is not so easy to formalize our sense to a definition. Mathematically, we can state that:

Definition 1.2.1 (The First Definition of Volume)**.** *Volume is a measure of regions in three-dimensional space.*

But I think we can extend the definition above with atomic theory:

Definition 1.2.2 (The Second Definition of Volume)**.** *Volume is a measure of the amount of space occupied by many small particles of* ***the same type*** *at equilibrium state.*

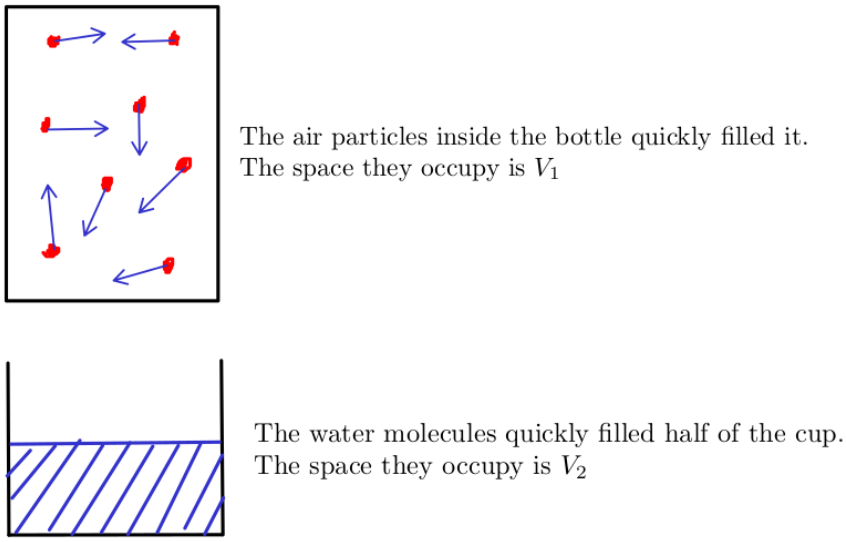

Figure 1.2: Definition of Volume

Now you might wonder if you mix 2 types of completely different gases, such as H_2 and He, are they the same type? The answer is yes, but until they reach the **equilibrium state**.

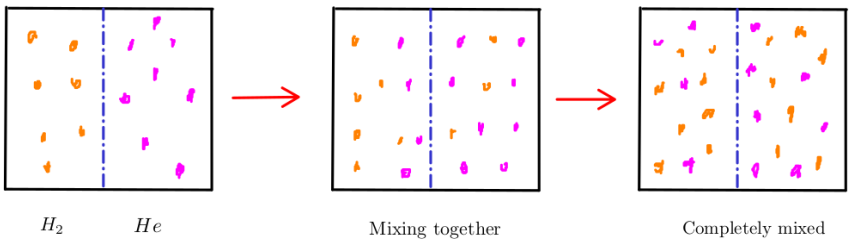

Figure 1.3: Mixing H_2 and He together

Reaching the equilibrium state means the density of gas mixture is the same at all locations inside the box.[1][2]

[1] This is just my claim.

[2] Mixing gas process can release an amount of heat energy. According to the zeroth law of thermodynamics, equilibrium state is reached when the temperature of the mixture is stable.

Unlike volume, **pressure** is very easy concept to define:

Definition 1.2.3 (Definition of Pressure)**.** *Pressure is the force applied* ***perpendicular*** *to the surface of an object per unit area over which that force is distributed:*

$$P = \frac{F}{A} \ (N/m^2), \ or \ (Pa)$$

One classic example of pressure is pressing your hand against a small piston filled with gas. If you press hard and fast enough, you can observe small sparks inside the piston. We will be back to explain the phenomenon later.

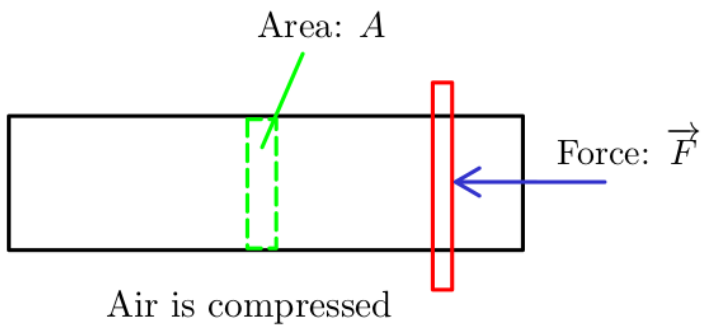

Figure 1.4: Pressing a small piston

After understanding what volume and pressure are, now it is time to answer our previous question from Chapter 0, what is Kelvin scale? Imagine you have an elastic balloon, which can be easily stretched. Of course, this balloon is filled fully with gas.

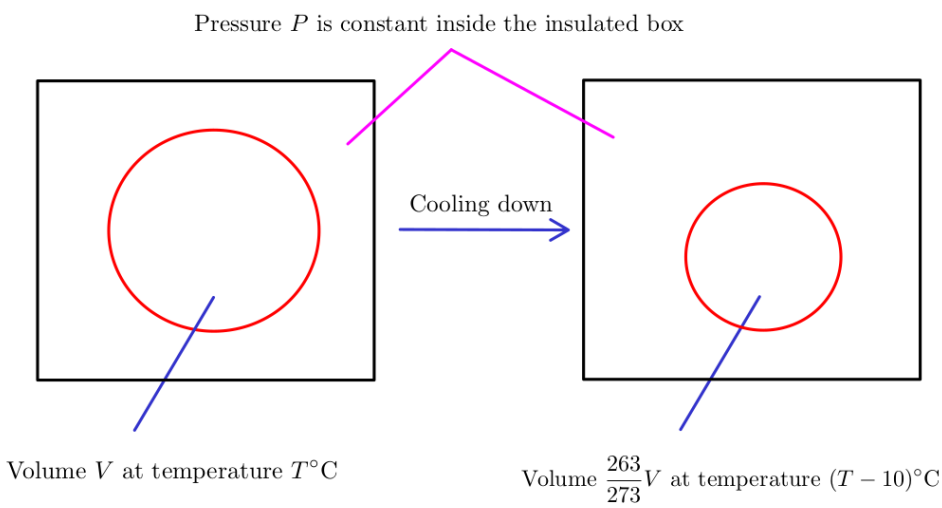

Figure 1.5: For every degree decrease, the volume $V/273$ decreases correspondingly

But because the volume quantity can not go below zero, if the temperature reaches $(T - 273)$°C, and continues to decrease, the volume will remain at zero. Alternatively, if you hold

the volume of the gas fixed, then its pressure will approach zero as the temperature approaches $(T - 273)°$C. The special value $-273°$C is called **absolute zero**[1], and defines the initial point of the **absolute temperature scale**, named after Lord Kelvin[2].

Definition 1.2.4 (Definition of Kelvin Scale)**.** *The Kelvin scale is an absolute temperature scale that starts at the lowest possible temperature, taken to be* $0K$*:*

$$T_K = T_C + 273°C \ (K)$$

1.3 The Ideal Gas Law

Discovered by experiments, many properties of a low-density gas can be summarized in the famous idea gas law, where P (N/m^2), V (m^3), n (mole), R is a universal constant, and T (K).

Theorem 1.3.1 (The Ideal Gas Law)**.** *The empirical ideal gas law for* ***low-density gas*** *is:*

$$PV = nRT$$

This section does not have much to discuss beyond introducing an important law of gases and some new constants. As you can see here, the unit of constant R is:

$$R = 8.31 \text{ (J/mol.K)}$$

If we define $N_A = 6.023.10^{23}$ (mol^{-1}) is Avogadro's number, then the law can be rewritten as:

$$PV = nRT = \frac{N}{N_A}RT = N\mathbf{k_B}T$$

The constant k_B is called **Boltzmann's constant**, it plays vital role in thermodynamics:

$$k_B = \frac{R}{N_A} = 1.381.10^{-23} \text{ (J/K)}$$

1.4 The Kinetic Theory of Gases

In a gas-filled piston, you already know that gas (or any substance) is simply a collection of tiny, chaotic, uncontrollable particles. If you heat the piston, these particles appear to move faster. In this section, we will use the gas model to quantify the relationship between piston temperature and the velocity of the particles.

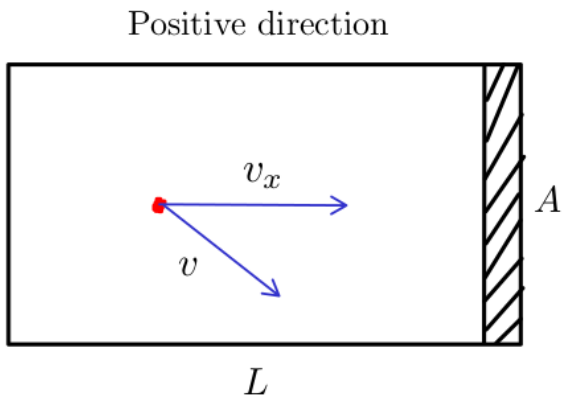

Figure 1.6: An oversimplified model of an ideal gas

[1]This is the basic idea of how absolute zero was discovered.
[2]How about holding both the volume and pressure of the gas fixed, and decrease temperature?

Since the system of 10^{23} particles is impossible to process in the first place, I have to make some assumptions here to simplify our physical model:

- My oversimplified model of an ideal gas contains only 1 particle.
- Its speed never changes.
- Piston is perfectly smooth, so collisions are always **elastic**.

Directly applying Definition 1.2.3 yields:

$$\begin{aligned}\overline{P}_{\text{x, on piston}} &= \frac{\overline{F}_{\text{x, on piston}}}{A} = -\frac{\overline{F}_{\text{x, on particle}}}{A} \text{ (Newton's Third Law)} \\ &= -\frac{m\dfrac{\Delta v}{\Delta t}}{A} = -\frac{m\dfrac{(-v_x) - v_x}{2L/v_x}}{A} = \frac{mv_x^2}{V}\end{aligned}$$

Or equivalent to:

$$\overline{P}_{\text{x, on piston}} V = mv_x^2$$

If our piston contains N particles, then the total pressure on piston by x-axis is:

$$P_x = N\overline{P}_{\text{x, on piston}} = \frac{Nmv_x^2}{V}$$